

Barenblatt Similarity Profiles for Every Positive Exponent: Exact Mass, Moment, Free-Boundary, and Dissipation Atlas

Route-A structural certificate C207

Abstract

For the one-dimensional equation $u_t = (u^m)_{xx}$, every $m > 0$ and mass $M > 0$, we classify the centered nonnegative integrable zero-flux first-kind similarity profiles. One derivation yields compact support for $m > 1$, the Gaussian at $m = 1$, and algebraic tails for $0 < m < 1$. Exact Beta integrals normalize mass and give every absolute moment, including the sharp threshold $r < (1+m)/(1-m)$ and the logarithmic second-moment boundary $m = 1/3$. We also give porous pressure/free-boundary data and a conditional rescaled dissipation identity. The statement does not classify arbitrary Cauchy solutions. Exact finite cells audit conventions but do not prove the continuous theorem. Every Route-A gate fails.

Keywords: porous-medium equation; fast diffusion; Barenblatt profile; similarity solution; moment threshold.

中文摘要

本文对一维方程 $u_t = (u^m)_{xx}$ 的全部 $m > 0$ 和给定质量, 分类中心化、非负、可积、零通量的第一类自相似剖面。统一推导覆盖紧支撑、高斯与代数尾三种区域, 并给出精确质量、全阶绝对矩、 $m = 1/3$ 二阶矩边界、压力自由边界及限定适用范围的重标度耗散。结论不扩张为任意初值问题解的分类; 路线A全部门失败。

1 Frozen class and profile equation

Let $m > 0$, $M > 0$, and consider

$$u_t = (u^m)_{xx}, \quad x \in \mathbb{R}, \quad t > 0. \quad (1)$$

We classify only centered, nonnegative, integrable first-kind profiles of mass M such that $F^m \in W_{\text{loc}}^{1,1}(\mathbb{R})$ and the integrated zero-flux law below holds almost everywhere. Uniqueness is up to almost-everywhere equality (equivalently, for the continuous representative determined by F^m). Translates, signed profiles, higher dimensions, and arbitrary Cauchy solutions are outside the claim. Source-type ownership and the surrounding theory are classical [1, 2]; no formula priority is claimed.

Mass-preserving scaling has

$$u(x, t) = t^{-\alpha} F(\xi), \quad \xi = xt^{-\alpha}, \quad \alpha = \frac{1}{m+1}.$$

Indeed exponent matching gives $\alpha + 1 = m\alpha + 2\alpha$. The profile equation (in distributions) and its zero-flux integral (almost everywhere) are

$$-\alpha(F + \xi F') = (F^m)'', \quad (F^m)' + \alpha \xi F = 0. \quad (2)$$

2 All positive exponents and exact mass

Theorem 1. *In the frozen class there is exactly one mass- M profile. For $m > 1$, set*

$$p = \frac{1}{m-1}, \quad k = \frac{m-1}{2m(m+1)}.$$

Then

$$F(\xi) = (C - k\xi^2)_+^p, \quad M = C^{p+1/2} k^{-1/2} B\left(\frac{1}{2}, p+1\right). \quad (3)$$

For $m = 1$,

$$F(\xi) = \frac{M}{2\sqrt{\pi}} e^{-\xi^2/4}. \quad (4)$$

For $0 < m < 1$, set

$$q = \frac{1}{1-m}, \quad b = \frac{1-m}{2m(m+1)}.$$

Then

$$F(\xi) = (C + b\xi^2)^{-q}, \quad M = C^{1/2-q} b^{-1/2} B\left(\frac{1}{2}, q - \frac{1}{2}\right). \quad (5)$$

In each nonlinear regime the displayed equation uniquely fixes $C > 0$.

Proof. Where $F > 0$, equation (2) gives

$$(F^{m-1})' = -\frac{\alpha(m-1)}{m} \xi \quad (m \neq 1), \quad (\log F)' = -\alpha \xi \quad (m = 1).$$

Integration on each positivity component gives (3)–(5). Local absolute continuity makes F continuous. In the porous case a finite endpoint forces $F^{m-1} = C - k\xi^2 = 0$; the two endpoints therefore have equal modulus, so the only nonempty component is $(-\sqrt{C/k}, \sqrt{C/k})$. In the fast and heat cases the displayed finite expressions cannot approach zero at a finite component endpoint, while integrability excludes a missing infinite end; their positivity set is all of $\mathbb{R}$. Thus no extra positivity component or zero interval is possible. Substituting $z = k\xi^2/C$ or $z = b\xi^2/C$ evaluates the two mass integrals by Euler's Beta integral; Gaussian integration gives (4). The porous mass is strictly increasing in C , while the fast mass is strictly decreasing because $q > 1$. Thus mass fixes C . Conversely the formulas satisfy (2), have mass M , and are centered. This proves existence and uniqueness inside the frozen profile class. $\square$

Revision focus. Round 0 freezes the profile class and proves the complete three-regime mass-normalized classification.

Declarations

Data and code availability. Package-local exact synthetic evidence and deterministic code accompany the paper; no observational data are used.

Ethics. No human, animal, personal, or sensitive data are used.

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References

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